```
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
import numpy as np

(x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load_data()

plt.imshow(x_train[1])

plt.imshow(x_train[0])

x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0
x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)

print(x_train.shape)
print(x_test.shape)
print(y_train.shape)
print(y_test.shape)

model = keras.Sequential([
    keras.layers.Conv2D(
        32,
        (3,3),
        activation='relu',
        input_shape=(28,28,1)
    ),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),
    keras.layers.Conv2D(
        64,
        (3,3),
        activation='relu'
    ),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),
    keras.layers.Conv2D(
        128,
        (3,3),
        activation='relu'
    ),
    keras.layers.Flatten(),
    keras.layers.Dense(128, activation='relu'),
    keras.layers.Dropout(0.25),
    keras.layers.Dense(10, activation='softmax')
])
```

```
model.summary()

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)


history = model.fit(
    x_train,
    y_train,
    epochs=10,
    validation_data=(x_test, y_test)
)


test_loss, test_acc = model.evaluate(
    x_test,
    y_test
)
print('Test accuracy:', test_acc)
```